

1 / Problem and delivered workflow

SpeechTrace: time-grounded speech delivery feedback

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An overall speaking score rarely tells a person what to listen to next. This prototype compares a short recording against a reference with the same transcript and connects measurable acoustic differences to specific moments.

The dashboard accepts reference audio, participant audio and a shared transcript, or one of 14 labeled comparisons. It displays a weighted reference-similarity score, aligned energy curves, detected regions, explanations, and estimated word timestamps. Clicking words or regions seeks both recordings to matched positions. Results export as JSON.

Bundled cases are clearly labeled Precomputed example. Reanalyze recordings runs the actual Python engine. Custom uploads always use the backend. Changed inputs invalidate results and abort outstanding analysis.

Live: <https://speechtrace-lab.vercel.app>

Code and complete dataset: <https://github.com/suhaslord/blob/tree/main/speechtrace>

Scope: English clips of 1–32 seconds and up to 120 transcript words. This is a reference-relative delivery tool. It does not score argument quality, speaking talent, or a universal ideal performance.

2 / Dataset construction and provenance

Two excerpts come from John F. Kennedy's January 20, 1961 inaugural address, credited to the John F. Kennedy Presidential Library & Museum. The downloaded source is public-domain government audio. Source offsets are 57.0 seconds (12-second Power excerpt) and 70.2 seconds (10.8-second Revolutionary Beliefs excerpt).

Exact source: archive.org/download/JFK_Inaugural_Address_19610120/JFK_Inaugural_Address_19610120.mp3

Source SHA-256: 7a931ad726a9b732d8db01af6eaff55b9a34d42ee78f7ea73eba8533c3aa0e4d

Rights evidence: commons.wikimedia.org/wiki/File:JFK_inaugural_address.ogg and archive.org/details/JohnF.KennedyInauguralAddress.

Each excerpt has seven paired cases: unchanged, global -10 dB gain control, near-perfect local -5 dB edit (3–5 s), moderate -14 dB edit (3–5 s), severe -24 dB edit (3–6 s), inserted two-second pause at 6 s, and a pitch-preserving 1.8x tempo mirror. These are digital edits, not newly recorded human performances.

Transcripts retain the original words. Transformation intervals are exact construction labels. Baseline word boundaries are model estimates, not manually verified word-level ground truth. The manifest includes audio hashes, labels, source offsets, estimated alignments and computed results.

All paired WAVs are checked in. Rebuilding requires Python and FFmpeg: `tools/download_source.py` then `tools/build_dataset.py`. `tools/evaluate.py` recomputes evaluation from the manifest.

3 / Acoustic and transcript alignment

Browser decoding converts supported audio into mono 16 kHz PCM WAV. The API validates sample rate, length, transcript length and audio energy before analysis. NumPy supplies deterministic signal processing; Vosk/Kaldi supplies pretrained English acoustic word decoding.

Frames are 25 ms long with 20 ms hops. A windowed FFT produces power spectra. A 26-band mel filterbank followed by a DCT yields 12 cepstral coefficients. Features also include RMS energy and autocorrelation-based pitch.

Cepstra are centered per recording. Energy is expressed relative to each recording's active-frame median. Pitch is expressed in semitones relative to its median voiced pitch. This removes overall offsets, but does not establish cross-speaker fairness.

Monotone dynamic time warping aligns cepstral and energy vectors at 100 ms resolution. The path maps participant time to reference time, permitting local pace and aligned energy comparisons without assuming equal duration.

Vosk decodes a constrained grammar based on the supplied transcript, with an unknown-token alternative. An ordered longest-common-subsequence match connects recognized words to the transcript and prevents spurious repeated words from misaligning later boundaries.

Coverage below 85% raises an uncertainty warning. Acoustic word times and confidence are reported rather than replaced with uniformly spaced fake timestamps. A pinned official model archive hash is verified before extraction to a temporary cache.

4 / Temporal explanations and scoring rubric

Energy: differences above 7 dB must last at least 400 ms. Pitch contour: differences above 5 relative semitones require at least 700 ms of voiced frames. Pacing: local acoustic path progression and matched speech-span ratios identify tempo shifts.

Pause candidates combine low-energy intervals with excess word gaps greater than 0.7 seconds. Word-gap evidence requires adjacent boundary confidence above 0.8. Explanations display interval, measured delta, units and overlapping transcript words.

Intervals that overlap within a component are merged. Each component score is $\max(0, 100 - 400 \times \text{flagged_duration} / \text{participant_duration})$. Overall score = $0.4 \times \text{energy} + 0.4 \times \text{pacing} + 0.2 \times \text{contour}$. The rubric is explicit and reproducible; it has not been calibrated against human competition judges.

A pause can affect both energy and pacing under this rubric. Detected boundaries can include neighboring natural silence. A changed contour or tempo can be intentional artistic delivery; the dashboard tells users to listen before changing their performance.

Causal scope: known transformations create measurable changes, and the engine reports the associated acoustic evidence. The prototype does not infer a speaker's intention, confidence, personality or other psychological cause.

5 / Evaluation and meaningful verification

The Power excerpt calibrates thresholds. Revolutionary Beliefs is a held-out excerpt from the same speaker. Evaluation comprises 14 pairs, two excerpts and one speaker; it is a small stress test, not a population benchmark.

Unchanged and global-gain controls: 0 false-positive cases out of four. Moderate and severe volume edits: mean interval intersection-over-union 0.975. Inserted pauses: mean interval IoU 0.815. Both 1.8x tempo mirrors produced pacing flags. Both near-perfect 5 dB local edits stayed below threshold.

Power scores: unchanged/gain/near-perfect 100; moderate 76; severe 60; pause 38.5; rushed 60. Beliefs scores: unchanged/gain/near-perfect 100; moderate 70.4; severe 60; pause 43.8; rushed 60.

Five automated engine tests cover controls, severity ordering, interval overlap, pauses, pacing, repeat-analysis consistency, and invalid/silent/wrong-rate input rejection. The submission audit records browser and production checks, including actual uploads, stale requests, seeking, exports, responsive layouts and the live API.

A real repeated-word alignment bug created a false pause in the Beliefs pause case; ordered matching fixed it. Invalid WAV EOF errors now return a useful 400 response. Independent upload preparation prevents one rapid file upload from silently cancelling the other.

Limitations: one speaker, short clips, English only, synthetic edits, no independently annotated word boundaries, no human-judge agreement study, and no cross-speaker evaluation. Silence and noise can distort normalization; uncertain alignment is shown.

6 / Deployment, privacy and reproducibility

Vercel hosts the static HTML/CSS/JavaScript dashboard and a Python `/api/analyze` function. `requirements.txt` pins NumPy and Vosk versions. No paid API keys, database, account system or Sites hosting are used.

First model use downloads and verifies the small official Vosk English model. Warm requests reuse its temporary cache. Cold starts can be slower; failures produce an error and allow bundled examples to remain usable.

Audio conversion happens in the browser. On Analyze, converted recordings and transcript are sent to the function for that request. Application code does not persist audio or transcripts and does not log request contents. Ordinary hosting request metadata still exists. The UI advises against confidential recordings.

Reproduce with Python 3.12: `pip install -r requirements.txt; python dev-server.py; python -m unittest qa/test_engine.py -v; python tools/evaluate.py`. FFmpeg is needed only to rebuild dataset variants. Dataset and model hashes are pinned, and the WAV examples are included.

AI disclosure: Codex assisted with scoping, implementation, debugging, transformations, tests, documentation and submission. Vosk/Kaldi is pretrained; no acoustic model was trained by this team. Thresholds were calibrated on one excerpt. Project code is MIT licensed; upstream dependencies and audio retain their own licenses and attribution.

The repository folder, manifest and technical report describe the same implementation. Source: github.com/suhaslord/blob/tree/main/speechtrace. Live evidence: speechtrace-lab.vercel.app/docs.html.